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University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

First Year Examination - Semester II - UCSC AY21 [held in March/April 2025]

IS1212 – Probability and Statistics

(Two (2) Hours)

Answer ALL questions

Number of Pages = 9 Number of Questions = 4

To be completed by the candidate

Index Number: | | | | | | | | |

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Important Instructions to candidates:

I. Students should answer in the medium of English language only using the space provided in this question paper.

II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.

III. Write your index number CLEARLY on each and every page of this Question paper.

IV. This paper consists of 4 questions in 9 pages (including the Cover Page).

V. Answer ALL questions.

VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.

VII. Non-Programmable calculators are allowed

VIII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

Question 1.

(a) What is the Sample Space of the random experiment where 5 fair coins are tossed simultaneously, and observe the number of heads appeared?

(b) What is the standard deviation of a Poisson distribution with mean 25?

(c) What is the mean of the Binomial distribution with n=10 and p=0.3?

(d) If a random variable X is normally distributed with mean 15 and variance 7, what is P(X=12)?

(e) If the random variable X ~N(30,3) and the random variable Y~N(20,7) what is the distribution of X-Y?

(f) What is the mean of a discrete uniform distribution which is defined in the range of (2,6)?

(g) Let $\bar{X}$ be the mean of a random sample of size 30 taken from a $N(150,60)$ distribution What is the distribution of $\bar{X}$?

(h) The mean of a binomial distribution with $n = 10$ is 2. Find the other parameter, probability of success (p) in this distribution.

(i) Suppose that the mean and the standard deviation of a binomial distributed random variable are 1.6 and 1.44 respectively. Find the parameters of this binomial distribution.

(j) If $X \sim N(60,36)$ distribution, determine the distribution of $3X$.

(25 Marks)

Question 2.

(a) The mean of 4 numbers is 5 and the mean of 3 different numbers is 12. What is the mean of all seven numbers together?

(b) If the mean of the numbers 12, 18, k, 21, 13 is 15, find the value of k.

(c) Let $P(A) = 0.3$, $P(B) = 0.2$, and $P(A \cup B) = 0.4$. Find $P(A \cap B)$?

(d) If the events A and B are independent and $P(A) = 0.3$, $P(B) = 0.2$, find $P(A \cup B)$?

(e) The probability distribution of the random variable X is given in the table below.

X	2	3	4
P(X=x)	3/8	1/2	1/8

Find E(X)

(f) Suppose that the weights of first-year university students have symmetrical, bell-shaped distribution with the mean of 60kg and the standard deviation of 36. Using the empirical rule in statistics, find the approximate percentage of weights between 60kg and 66 kg.

(g) If $P(A|B)=2/5$, $P(B) = 1/4$, $P(A) =1/3$ find $P(A \cap B)$.

(h) If $P(A|B)=2/5$, $P(B) = 1/4$, $P(A) =1/3$ find $P(B|A)$.

(i) In a race, the probability that Pathum wins is 1/3, the probability that Sithum wins is 1/4, and the probability that Nethum wins is 1/5. Find the probability that Pathum or Nethum wins the race.

(j) A die is rolled twice. Find the probability of obtaining a 4 on the first roll and an odd number on the second roll.

(25 Marks)

Question 3

(a) Suppose a boy hits a dartboard 10 number of times and the probability that the dart is thrown directly at the center of the dartboard is 0.3. Let X be the number of times that the dart hits **outside** the center of the dartboard.

(i) Write down the distribution of the random variable X.

(3 Marks)

(ii) What is the maximum value that X can take?

(2 Marks)

(iii) Find the probability that exactly two hits at the center of the dartboard.

(5 Marks)

(b) If a random variable X has a Poisson distribution with mean rate λ, it has the probability distribution given by $\frac{e^{-\lambda}\lambda^{x}}{x!}$.

(i) An insurance company receives on average 2 claims per week from a certain factory. Assuming that the number of claims follows a Poisson distribution, find the probability that it receives more than 1 claim in a given fortnight (two weeks).

(5 Marks)

(ii) If a random sample of size 30 is taken from Poisson(4.5) distribution, what is the probability that sample mean exceeds 5?

(5 Marks)

(c) Suppose that a patient is waiting for a suitable blood donor and the probability that the selected donor will be a match is 0.2. Calculate the probability that the first matched donor is the fifth donor tested for matching.

(5 Marks)

Question 4

(a) A certain type of cabbage has a mass which is normally distributed with a mean of 1kg and a standard deviation of 0.15kg. In a lorry load of 500 of these cabbages, estimate how many will have mass;

(i) less than 1.2kg?

(5 Marks)

(ii) greater than 0.75kg?

(5 Marks)

(b) John washes and polishes cars. The time he takes to wash a car is a normally distributed random variable with a mean of 15 minutes and a standard deviation of 3 minutes. The time he spends on polishing a car is a normal random variable with a mean of 20 minutes and a standard deviation of 5 minutes. Find the probability that

(i) he spends more than 14 minutes to wash a randomly selected car.

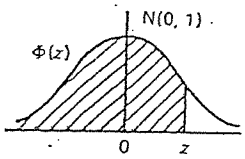
(5 Marks)

(ii) he spends more than 40 minutes to complete a job, which means wash and polish a car.

(5 Marks)

(iii) he spends less time on washing a car than polishing.

(5 Marks)



THE DISTRIBUTION FUNCTION $\Phi(z)$ OF THE NORMAL DISTRIBUTION N(0, 1)

z	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
															ADD				
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359	4	8	12	16	20	24	28	32	36
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753	4	8	12	16	20	24	28	32	36
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141	4	8	12	15	19	23	27	31	35
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517	4	7	11	15	19	22	26	30	34
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879	4	7	11	14	18	22	25	29	32
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224	3	7	10	14	17	20	24	27	31
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549	3	7	10	13	16	19	23	26	29
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852	3	6	9	12	15	18	21	24	27
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133	3	5	8	11	14	16	19	22	25
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389	3	5	8	10	13	15	18	20	23
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621	2	5	7	9	12	14	16	19	21
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830	2	4	6	8	10	12	14	16	18
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015	2	4	6	7	9	11	13	15	17
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177	2	3	5	6	8	10	11	13	14
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319	1	3	4	6	7	8	10	11	13
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441	1	2	4	5	6	7	8	10	11
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545	1	2	3	4	5	6	7	8	9
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633	1	2	3	4	4	5	6	7	8
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706	1	1	2	3	4	4	5	6	6
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767	1	1	2	2	3	4	4	5	5
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817	0	1	1	2	2	3	3	4	4
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857	0	1	1	2	2	2	3	3	4
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890	0	1	1	1	2	2	2	3	3
2.3	.9893	.9896	.9898								0	1	1	1	1	2	2	2	2
				.9901	.99036	.99061	.99086				3	5	8	10	13	15	18	20	23
								.99111	.99134	.99158	2	5	7	9	12	14	16	18	21
2.4	.99180	.99202	.99224	.99245	.99266						2	4	6	8	11	13	15	17	19
						.99286	.99305	.99324	.99343	.99361	2	4	6	7	9	11	13	15	17
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520	2	3	5	6	8	9	11	12	14
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643	1	2	3	5	6	7	8	9	10
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736	1	2	3	4	5	6	7	8	9
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807	1	1	2	3	4	4	5	6	6
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861	0	1	1	2	2	3	3	4	4
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900	0	1	1	2	2	2	3	3	4
3.1	$.9^{3}032$	$.9^{3}065$	$.9^{3}096$								3	6	9	13	16	19	22	25	28
				$.9^{3}126$	$.9^{3}155$	$.9^{3}184$	$.9^{3}211$				3	6	8	11	14	17	20	22	25
								$.9^{3}238$	$.9^{3}264$	$.9^{3}289$	2	5	7	10	12	15	17	20	22
3.2	$.9^{3}313$	$.9^{3}336$	$.9^{3}359$	$.9^{3}381$	$.9^{3}402$						2	4	7	9	11	13	15	18	20
						$.9^{3}423$	$.9^{3}443$	$.9^{3}462$	$.9^{3}481$	$.9^{3}499$	2	4	6	8	9	11	13	15	17
3.3	$.9^{3}517$	$.9^{3}534$	$.9^{3}550$	$.9^{3}566$	$.9^{3}581$						2	3	5	6	8	10	11	13	14
						$.9^{3}596$	$.9^{3}610$	$.9^{3}624$	$.9^{3}638$	$.9^{3}651$	1	3	4	5	7	8	9	10	12
3.4	$.9^{3}663$	$.9^{3}675$	$.9^{3}687$	$.9^{3}698$	$.9^{3}709$	$.9^{3}720$	$.9^{3}730$	$.9^{3}740$	$.9^{3}749$	$.9^{3}758$	1	2	3	4	5	6	7	8	9
3.5	$.9^{3}767$	$.9^{3}776$	$.9^{3}784$	$.9^{3}792$	$.9^{3}800$	$.9^{3}807$	$.9^{3}815$	$.9^{3}822$	$.9^{3}828$	$.9^{3}835$	1	1	2	3	4	4	5	6	7
3.6	$.9^{3}841$	$.9^{3}847$	$.9^{3}853$	$.9^{3}858$	$.9^{3}864$	$.9^{3}869$	$.9^{3}874$	$.9^{3}879$	$.9^{3}883$	$.9^{3}888$	0	1	1	2	2	3	3	4	5
3.7	$.9^{3}892$	$.9^{3}896$	$.9^{3}900$	$.9^{4}04$	$.9^{4}08$	$.9^{4}12$	$.9^{4}15$	$.9^{4}18$	$.9^{4}22$	$.9^{4}250$									
3.8	$.9^{4}28$	$.9^{4}31$	$.9^{4}33$	$.9^{4}36$	$.9^{4}38$	$.9^{4}41$	$.9^{4}43$	$.9^{4}46$	$.9^{4}48$	$.9^{4}500$									
3.9	$.9^{4}52$	$.9^{4}54$	$.9^{4}56$	$.9^{4}58$	$.9^{4}59$	$.9^{4}61$	$.9^{4}63$	$.9^{4}64$	$.9^{4}66$	$.9^{4}670$									

For negative values of z use $\Phi(z) = 1 - \Phi(-z)$

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